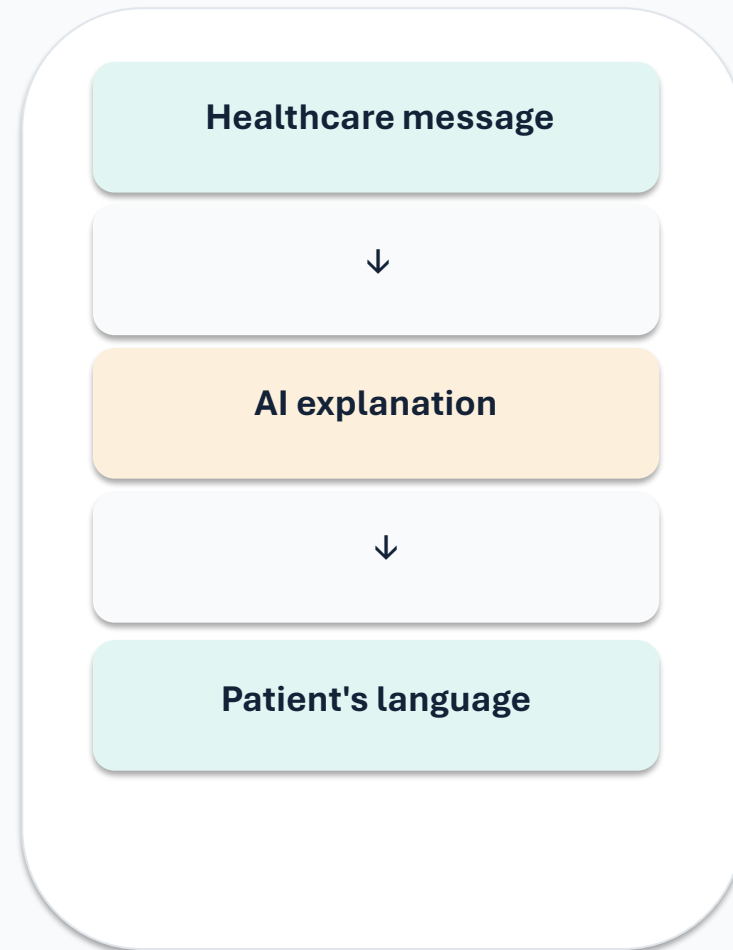


Healthcare Easy

Translate • Explain • Simplify



Making healthcare information easier to understand across language barriers.



The problem

02

A patient may be able to read...

...but still struggle with medical terminology, instructions, or what action they are expected to take.

Health Literacy in the Netherlands

52.8%

struggle with health information

Language

Limited understanding can create an additional barrier.

Complex terminology

Translation alone may still leave medical language **difficult** to understand.

Action & meaning

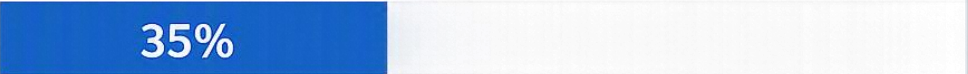
People may struggle with what they need to do.

SDG – Reduced Inequalities

03

Access to understandable healthcare information can be more difficult for some groups than for others.

Limited Health Literacy in the Netherlands



35%

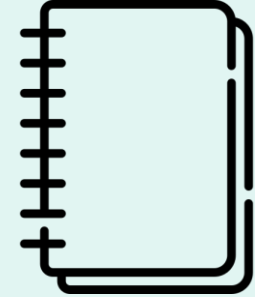
35% of adults have limited health literacy.

Relevance

Unequal access to understandable healthcare information.

User Group

People in the Netherlands who have difficulty understanding Dutch healthcare information because of language or medical terminology.



Primary

Patients with limited Dutch proficiency.

Secondary

Patients who understand Dutch but struggle with complex medical terminology.

Who is not the intended user

Healthcare workers or medical emergencies requiring immediate intervention.

The solution

05

1

Source language

Dutch / Arabic / English etc.

2

Healthcare message

Paste the received text

3

AI

Structured + safety-focused

4

Target language

Arabic / English /
Romanian /
Lithuanian /
Dutch

Simple explanation

What the message means in
clear language.

Actions

Only actions explicitly stated in
the original.

Important / unclear

Important facts + information
that is missing or unclear.

Safety note

No diagnosis or treatment
advice; human support when
needed.

What we built

A working Python MVP powered by Gemini.

User flow

1. Choose source language
2. Choose target language
3. Paste a multi-line healthcare message
4. Confirm the request
5. Receive a structured explanation

Technology

-  Python
-  Google Gemini API
- ✓ Pydantic structured output
- ▣ Rich terminal interface
- عربي Arabic text rendering with arabic-reshaper + python-bidi

Supported languages

Dutch
English
Arabic
Romanian
Lithuanian

Important implementation

All AI-generated result fields use the selected target language, not only the final explanation.

Security choice

The Gemini API key is stored as an environment variable, not hard-coded in the Python file.

Demo

Safety by design

06

Healthcare AI must explain, not invent.

The original message is the source of truth

Preserve medication names, dosages, numbers, dates, times, warnings and instructions.

The AI must not

Diagnose • prescribe • create treatment plans • guess missing information • change medical facts.

Privacy

Healthcare information may contain sensitive data. Users should avoid unnecessary personal information.

Transparency

The user should know the explanation is generated by AI.

Human support

The tool does not replace doctors, pharmacists or professional interpreters.

When information is unclear

Important medical decisions should be checked with a healthcare professional.

Impact, MVP & next step

**Unequal access to understandable healthcare information →
a practical problem our MVP explores.**

What is working now

- ✓ Live Gemini request
- ✓ 5 languages
- ✓ Structured output
- ✓ Multi-line input
- ✓ Target-language output
- ✓ Safety constraints

The goal

Help a person understand a specific healthcare message without changing its important meaning — and know when to seek human help.